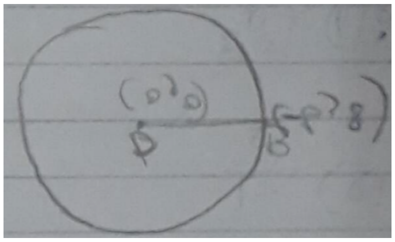
<page_start no = 1>
<sol_start id=1>
1. $p(x) = x^2-2x-(7p+3)$
$P(-4) = 16-2(-4)-(7P+3)$
$= 16 + 8-7P-3$
$O = 24-3-7P$
$= 21 -7P$
$7P = 21$
$P = 3$
(1) 3
<sol_end>

<sol_start id=2>
2. (4) No solution
<sol_end>

<sol_start id=3>
3. Radius = OB



$$OB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(6-0)^2 + (8-0)^2}$$

$$= \sqrt{6^2 + 8^2}$$

$$= \sqrt{36+64}$$

$$= \sqrt{100}$$

= 10 units
(1) 10 units
<sol_end>

<sol_start id=4>
4. (2) Proportional
<sol_end>

<sol_start id=5>
5. $\sqrt{3}\tan\theta = 2\sin\theta$

$$\sqrt{3}\,\frac{\cancel{\sin\theta}}{\cos\theta} = 2\cancel{\sin\theta}$$

$$\frac{\sqrt{3}\cdot 1}{2\cos\theta}$$

$$\frac{\sqrt{3}}{2} = \cos\theta$$

$$\theta = 30^\circ$$

$$\sin^2\theta - \cos^2\theta$$

$$= \sin^2 30^\circ - \cos^2 30^\circ$$

$$= \left(\frac{1}{2}\right)^2 - \left(\frac{\sqrt{3}}{2}\right)^2$$

$$= \frac{1}{4} - \frac{3}{4}$$

$$= \frac{-2}{4} = \frac{-1}{2}$$

(2) $\frac{-1}{2}$

<sol_end>
<page_end>

<page_start no = 2>
<sol_start id=6>
6. (1) x^3y^3
<sol_end>

<sol_start id=7>
7. $4x^2-2x+(k-4)$

$$\alpha + \beta = \frac{1}{2}\alpha\beta$$

$$\alpha + \beta = \frac{2}{4}\left(\frac{-b}{a}\right)$$

$$= \frac{1}{2}$$

$$\alpha\beta = \frac{c}{a}$$

$$= \frac{k-4}{4}$$

$$\alpha + \beta = \frac{1}{2}\alpha\beta$$

$$\frac{1}{2} = \frac{1}{2}\left(\frac{k-4}{4}\right)$$

$$\frac{1}{2} = \frac{k-4}{8}$$

$$\frac{8}{2} = k-4$$

k–4 = 4
k = 4 + 4
= 8

<sol_end>

<sol_start id=8>

8. A(0, 0) ; B(3, $\sqrt{3}$); c (3, x)
AB = BC = CA

$$AB = \sqrt{\left(x_2 - x_1\right)^2 + \left(y_2 - y_1\right)^2}$$

$$= \sqrt{\left(3-0\right)^2 + \left(\sqrt{3}-0\right)^2}$$

$$= \sqrt{3^2 + 3}$$

$$= \sqrt{9+3}$$

$$= \sqrt{12}$$

$$BC = \sqrt{\left(3-3\right)^2 + \left(x\sqrt{3}\right)^2}$$

$$\sqrt{2} = \sqrt{0 + x^2 - 2\sqrt{3}x + 3}$$

$$x^2 - 2\sqrt{3}x + 3 = 12$$

$$x^2 - 2\sqrt{3}x - 9 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{2\sqrt{3} \pm \sqrt{\left(2\sqrt{3}\right)^2 - 4(1)(-9)}}{2}$$

$$= \frac{2\sqrt{3} \pm \sqrt{12+36}}{2}$$

$$= \frac{2\sqrt{3} \pm \sqrt{48}}{2}$$

$$= \frac{2\sqrt{3} \pm 4\sqrt{3}}{2}$$

$$= \frac{2\sqrt{3} - 4\sqrt{3}}{2} \quad ; \quad \frac{2\sqrt{3} + 4\sqrt{3}}{2}$$

$$= \frac{-\cancel{2}\sqrt{3}}{\cancel{2}} \quad ; \quad \frac{-\cancel{6}\sqrt{3}}{\cancel{2}}$$

$$= -\sqrt{3} \quad ; \quad = 3\sqrt{3}$$

$x = -\sqrt{3}$ is rejected

$x = 3\sqrt{3}$

<sol_end>
<page_end>

<page_start no = 3>
<sol_start id=9>

9. $2\,px + 3y = 7$

$2x + y = 6$

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \Rightarrow \text{ one solution}$$

$$-\frac{2p}{2} \neq \frac{3}{1}$$

$$p \neq 3$$

<sol_end>

<sol_start id=10>

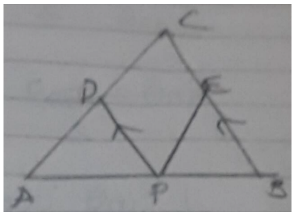
10. given

PD // BC

$$\frac{AD}{PC}=\frac{CE}{BE}$$

To prove that

EP // AC



Statement	Reason
PD // CB	Given
$\frac{AD}{DC}=\frac{AP}{PB}$ (1)	Basic proportionality theorem
$\frac{AD}{DC}=\frac{CE}{BE}$ (2)	given
From (1) and (2)	
$\frac{AP}{PB}=\frac{CE}{BE}$	
since $\frac{AP}{PB}=\frac{CE}{BE}$	Converse of Basic Proportionality theorem
EP // AC	

<sol_end>

<page_end>

<page_start no = 4>

<sol_start id=11>

11. $\tan\theta=\frac{1}{2}$

$$\frac{\sin\theta}{\cos\theta}=\frac{1}{2}$$

$$\sin\theta=\frac{\cos\theta}{2}$$

$\frac{1}{2}\sin\theta$ $\qquad$ $\cos\theta=\frac{1}{5}$

$$\frac{1}{2}\times\frac{\cos\theta}{2}\times\cos\theta=\frac{1}{5}$$

$$\frac{\cos^2\theta}{4}=\frac{1}{5}$$

$$\cos^2\theta=\frac{4}{5}$$

$$\cos\theta=\frac{2}{\sqrt{5}}$$

$$\sin\theta=\frac{\cancel{2}}{\sqrt{5}}\times\frac{1}{\cancel{2}}$$

$$=\frac{1}{\sqrt{5}}$$

<sol_end>

<sol_start id=12>

12. $5^4\times2^3 = 5000$

$n = 3$

$m = 4$

$$\frac{257\times2}{5^4\times2^4}$$

$$=\frac{.514}{10,000}$$

$= 0.0514$

<sol_end>

<sol_start id=13>

13. assume = $\sqrt{3}$ is rational.

$\sqrt{3}=\frac{a}{b}$ (a and b are coprime)

$$3=\frac{a^2}{b^2}$$

$3b^2 = a^2$

$a = 3C$ …..(1)

$(3c)^2 = 3b^2$

$\cancel{9}c^2 = \cancel{3}b^2$ …..(2)

b is divisible by 3

Since a and b are divisible by 3.

They are not co-prime. Hence $\sqrt{3}$.

an irrational number.

<sol_end>

<sol_start id=14>

14. $p(x) = 2x^2 - 3x + 7$

$$\alpha+\beta=\frac{3}{2}$$

$$\alpha\beta=\frac{7}{2}$$

$$\frac{1}{2\alpha-3}+\frac{1}{2\beta-3}$$

$$\frac{2\beta-3+2\alpha-3}{4\alpha\beta-6\alpha-6\beta+9}$$

$$=\frac{2(\alpha+\beta)-6}{-2(3\alpha+3\beta-4\alpha\beta)+9}$$

$$=\frac{\cancel{2}\left(\frac{3}{\cancel{2}}\right)-6}{-2\left(3(\alpha+\beta)-\cancel{4}\left(\frac{7}{\cancel{2}}\right)\right)+9}$$

$$=\frac{3-6}{-2\left(3\left(\frac{3}{2}\right)-14\right)+9}$$

$$=\frac{-3}{-\cancel{2}\left(\frac{9-28}{\cancel{2}}\right)+9}$$

$$=\frac{-3}{-\cancel{9}+28+\cancel{9}} = -3/28$$

$3x - (a + 1)y - (2b-1)$

5x + (1–2a)y – 3b

$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \Rightarrow$ infinitely many solutions.

$$\frac{3}{5} = \frac{-a-1}{1-2a} = \frac{2b-1}{3b}$$

$$\frac{-a-1}{1-2a} = \frac{3}{5}$$

–5(a+1)=3 (1–2a)

–5a –5 = 3–ba

–5a + ba = 3+5

a = 8

$$\frac{2b-1}{3b} = \frac{3}{5}$$

5(2b–1) = 3(3b)

10b–5 = 9b

10b–9b = 5

b = 5

<sol_end>

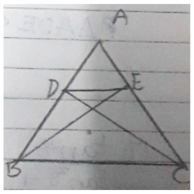
<sol_start id=16>

16. Given

$\Delta ABE \cong \Delta ACD$

To prove that

$\Delta ADE \sim \Delta ABC$



Statement	Reason
$\frac{AD}{DB} = \frac{AE}{EC}$	Basic proportionality theorem
DE // BC	
$\angle A = \angle A$	Common angle
$\Delta ADE \sim \Delta ABC$	By SAS similarity criteria.

<sol_end>

<sol_start id=17>

17. $$\frac{\tan^2\theta}{\tan^2\theta-1} + \frac{\cos ec^2\theta}{\sec^2\theta-\cos ec^2\theta} = \frac{1}{\sin^2\theta-\cos^2\theta}$$

$$\frac{\sin^2\theta\times 1}{\cos^2\times\frac{\sin^2\theta-1}{\cos^2\theta}} + \frac{1\times 1}{\sin^2\theta\times\frac{1}{\cos\theta}-\frac{1}{\sin^2\theta}}$$

$$\frac{\sin^2\theta}{\cos^2\theta\times\frac{\sin^2\theta-\cos^2\theta}{\cos^2\theta}} + \frac{1}{\frac{\sin^2\theta\times\sin^2\theta-\cos^2\theta}{\sin^2\theta\cos^2\theta}}$$

$$\frac{\sin^2\theta+\cos^2\theta}{\sin^2\theta-\cos^2\theta}$$

$$\frac{1}{\sin^2\theta-\cos^2\theta} = \frac{1}{\sin^2\theta-\cos^2\theta}$$

LHS = RHS

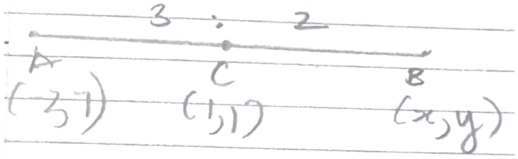
<sol_end>

<page_end>

<page_start no = 5>

<sol_start id=18>

18.



$$B = \left(\frac{x_2m_1+x_1m_2}{m_1+m_2};\frac{y_2m_1+y_1m_2}{m_1+m_2}\right)$$

$$= \left(\frac{1\times 3+(-2)2}{3+2};\frac{1\times 3+7\times 2}{3+2}\right)$$

$$= \left(\frac{3-4}{5};\frac{3+14}{5}\right)$$

$$= \left(\frac{-1}{5};\frac{m}{5}\right)$$

<sol_end>

<sol_start id=19>

19. $$\frac{2x}{y+1}=1$$

$2x = y+1$

$2x-y = 1 \qquad \ldots(1)$

$$\frac{x+4}{2y}=\frac{1}{2}$$

$$x+4=\frac{\cancel{2}}{\cancel{2}}y$$

$x-y = -4 \qquad \ldots(2)$

$$2x-\cancel{y}=1$$

$$x-\cancel{y}=-4$$

$x = 5$

$-y = -4 -5$

$= -9$

$y = 9$

$$\frac{x}{y}=\frac{5}{9}$$

<sol_end>

<sol_start id=21>

21. $$\frac{\sin 30^\circ + 2\cos 45^\circ + \tan^2 60^\circ}{\frac{1}{2}\times\cot 45^\circ + \cos^2 30^\circ + \tan^2 45^\circ}$$

$$=\frac{\frac{1}{2}+\cancel{2}\times\frac{1}{\cancel{\sqrt{2}}}+\sqrt{3}}{\frac{1}{2}\times 1+\left(\frac{\sqrt{3}}{2}\right)^2+1^2}$$

$$=\frac{\frac{1+2\sqrt{2}+2\sqrt{3}}{2}}{\frac{1}{2}+\frac{3}{4}+1} \quad =\frac{\frac{1+2\sqrt{2}+2\sqrt{3}}{\cancel{2}}}{\frac{2+3+4}{\cancel{4}}}$$

$$\frac{\left(1+2\sqrt{2}+2\sqrt{3}\right)\times 2}{9}$$

$$=\frac{2+4\sqrt{2}+4\sqrt{3}}{9}$$

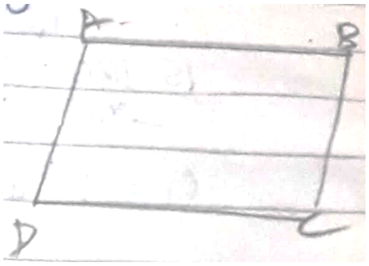
<sol_end>

<sol_start id=22>

22. A(1, 2); B(2, 3); C(6, 5); D(x, y)

AB = DC

AD = BC



$$AB=\sqrt{\left(x_2-x_1\right)^2+\left(y_2-y_1\right)^2}$$

$$=\sqrt{\left(2-1\right)^2+\left(3-2\right)^2}$$

$$=\sqrt{1+1}$$

$$=\sqrt{2}$$

$$DC=\sqrt{\left(x-6\right)^2+\left(y-5\right)^2}$$

$$=\sqrt{x^2-12x+36+y^2-10y+25}$$

$$\left(\sqrt{2}\right)^2=\left(\sqrt{x^2+y^2-12x-10y+61}\right)^2$$

$2 = x^2 + y^2 - 12x - 10y + 61$

$-59 = x^2 - 12x + y^2 - 10y$ …(1)

BC = $\sqrt{\left(x_2-x_1\right)^2+\left(y_2-y_1\right)^2}$

$$=\sqrt{\left(6-2\right)^2+\left(5-3\right)^2}$$

$$=\sqrt{4^2+2^2}$$

$$=\sqrt{16+4}$$

$$=\sqrt{20}$$

AD = $=\sqrt{\left(x-1\right)^2+\left(y-2\right)^2}$

$$\left(\sqrt{20}\right)^2 = \left(\sqrt{x^2 - 2x + 1 + y^2 - 4y + 4}\right)^2$$

$20 = x^2 - 2x + y^2 - 4y + 5$ …(2)

$15 = x^2-2x+y^2-4y$

$-59 = x2-12x+y^2-10y$

$= x(x-12)+y(y-10)$

$-5+ = (x+y)\ )\ (x-12)$

$x = 12$ $x = -y$

$x = 12$

$y = -12$

<sol_end>

<page_end>

<page_start no = 6>

<sol_start id=23>

23. Circumference = 1980 m

A = 330 m/min

B = 198 m/min

C = 220 m/min

D 1980

$$S = \frac{D}{T}$$

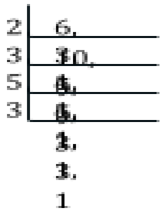
$$T = \frac{D}{S}$$

$$T_1 = \frac{198\cancel{0}}{33\cancel{0}} = 6\,\text{min}$$

$$T_2 = \frac{\cancel{198}0}{\cancel{198}} = 10\,\text{min}$$

$$T_3 = \frac{198\cancel{0}}{22\cancel{0}} = 9\,\text{min}$$

HCF of 6, 10, 9



2	6,
3	[illegible]
5	[illegible]
3	[illegible]
	[illegible]
	[illegible]
	1

= 90 mins = They will meet after 90 mins at starting point.

<sol_end>

<sol_start id=24>

24. $2x + y = 70$ y = Father

$2y + x = 95$ x = Son

$$\cancel{x+2y=95} \qquad \cancel{2x}+4y=190$$

$$\cancel{(2x+y=70)}\times 2 \qquad \cancel{2x}+y=70$$

$3y = 120$

$y = 40$

$x = 15$

Father = 40, Son = 15

<sol_end>

<page_end>

<page_start no = 7>

<sol_start id=26>

26. $\operatorname{cosec}\theta - \sin\theta = m$

$\sec\theta - \cos\theta = n$

$$\frac{1}{\sin\theta} - \sin\theta = m$$

$$\frac{1-\sin^2\theta}{\sin\theta} = m$$

$$\frac{\cos^2\theta}{\sin\theta} = m$$

$$\frac{1}{\cos\theta} - \cos\theta = n$$

$$\frac{1-\cos^2\theta}{\cos\theta} = n \quad n = \frac{\sin^2\theta}{\cos\theta}$$

$$\left(m^2n\right)^{\frac{2}{3}}+\left(mn^2\right)^{\frac{2}{3}}$$

$$\left(\frac{\cos^{\cancel{4}}\theta}{\cancel{\sin^2\theta}}\times\frac{\cancel{\sin^2\theta}}{\cancel{\cos\theta}}\right)^{\frac{2}{3}}+\left(\frac{\cancel{\cos^2\theta}}{\cancel{\sin\theta}}\times\frac{\sin^{\cancel{4}}\theta}{\cancel{\cos^2\theta}}\right)^{\cancel{\frac{2}{3}}}$$

$$\cos^{\cancel{3}}\theta^{\frac{2}{\cancel{3}}}+\sin^{\cancel{3}\times\frac{2}{\cancel{3}}}$$

$$\cos^2\theta+\sin^2\theta$$

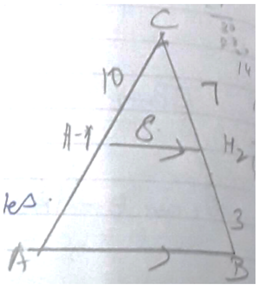
1 = 1

LHS = RHS

Hence proved

<sol_end>

<sol_start id=27>



<sub_id = 1>

27. (ii) 4.26 miles

<sub_id end>

<sub_id = 2>

(iii) 18.2

<sub_id end>

<sub_id = 3>

(iv) Hotel B to mountain = 5 miles

3x + 7x = 5

10x = 5

$$x=\frac{5}{10}$$

$$=\frac{1}{2}$$

$$\frac{7}{2}=3.5\,\text{miles}$$

<sub_id end>

<sol_end>

<sol_start id=28>

<sub_id = 1>

28. (ii) PR = $\sqrt{(18-2)^2+(3-5)^2}$

$=\sqrt{6^2+2^2}$

$=\sqrt{36+4}$

$=\sqrt{40}$

<sub_id end>
<sub_id = 2>

(iii) (3, 4)

$\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}\right)$

$=\left(\frac{10}{2},\frac{8}{2}\right)$

= (5, 4)

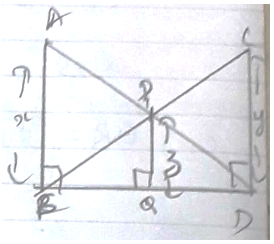
<sub_id end>
<sol_end>
<page_end>

<page_start no = 8>
<sol_start id=30>

30. Given $\underline{|ABD}=\underline{|CDB}=\underline{|PQB}=90°$

To prove that

$\frac{1}{x}+\frac{1}{y}=\frac{1}{z}$



Statement	Reason
$\underline{|ABD}=\underline{|CDB}=90°(R)$	Given
CD = CD (S)	Common side
AD = CB (H)	RHS
$\Delta ABD \cong CBD$	RHS $\cong$
x= y	
$\frac{1}{x}+\frac{1}{y}=\frac{1}{z}$	

<sol_end>

<page_end>